

YOUR 24/7 HEALTHCARE CHATBOT

A Voice-First Solution to Unlock HA Go for
Hong Kong's Older Adults

2025 Fall LANG 1406

Business Proposal Assignment

CHAN Lok Yin Matthew

SID: 21301380

Executive Summary

Hong Kong's rapidly aging population, projected to be over 25% aged 65+ by 2030 (Lau & Fong, 2021), faces exclusion from the digital healthcare system due to physical limitations, complex interfaces and cognitive decline. This creates a widening communication gap between seniors and essential healthcare services.

We propose 'Your 24/7 Healthcare Chatbot'—a voice-first AI assistant integrated into HA Go. Using Cantonese voice commands and large fonts, it allows seniors to manage appointments, medications, and health records conversationally, eliminating complex navigation.

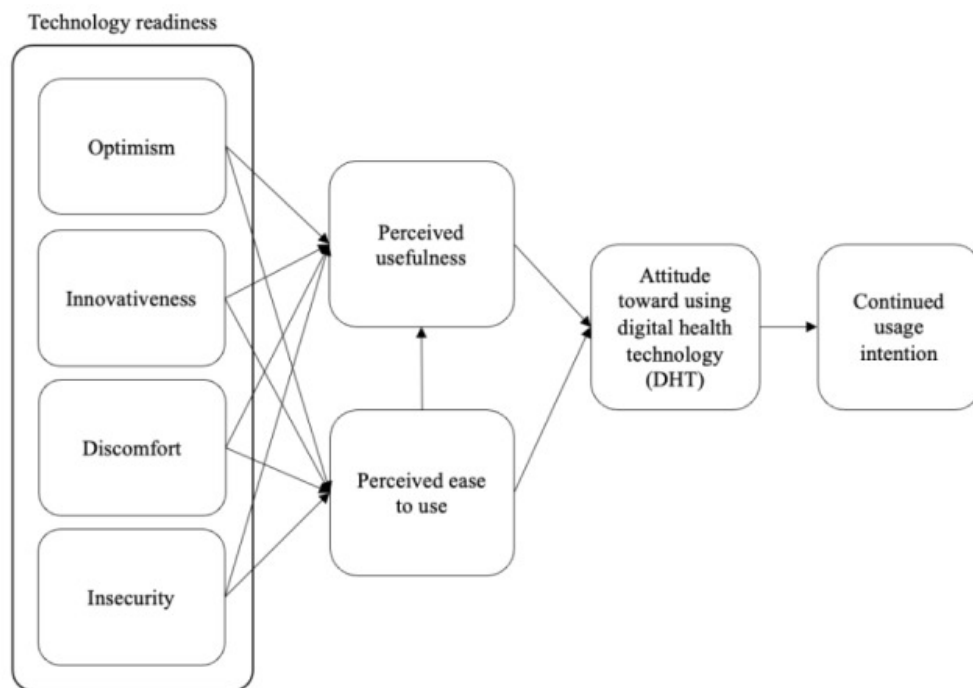
The solution directly increases perceived usefulness and ease of use while reducing the discomfort and insecurity that currently bar adoption. Early evidence from similar elderly-focused chatbots and positive senior testimonials confirms high acceptance potential. A partnership with the Hospital Authority will ensure encrypted data compliance and rapid integration.

We request HK\$2000000 for an 18-month pilot-to-full rollout. This modest investment will restore independence for thousands of older adults while reducing caregiver and clinic burden, delivering both social impact and measurable healthcare efficiency gains.

Introduction

Hong Kong's healthcare system has moved aggressively online, with the HA Go app becoming the central gateway for appointments, prescriptions, and records. For most users, this is convenient. But for many older adults, it is a daily barrier.

This proposal applies the Technology Readiness and Acceptance Model (TRAM) to diagnose the barriers seniors face with HA Go. By targeting the inhibitors that lower perceived usefulness and ease of use, we present “Your 24/7 Healthcare Chatbot” — an AI conversational assistant designed for Hong Kong's older adults. The principal objective is to transform HA Go from a source of frustration into an empowering, voice-driven lifelong companion.

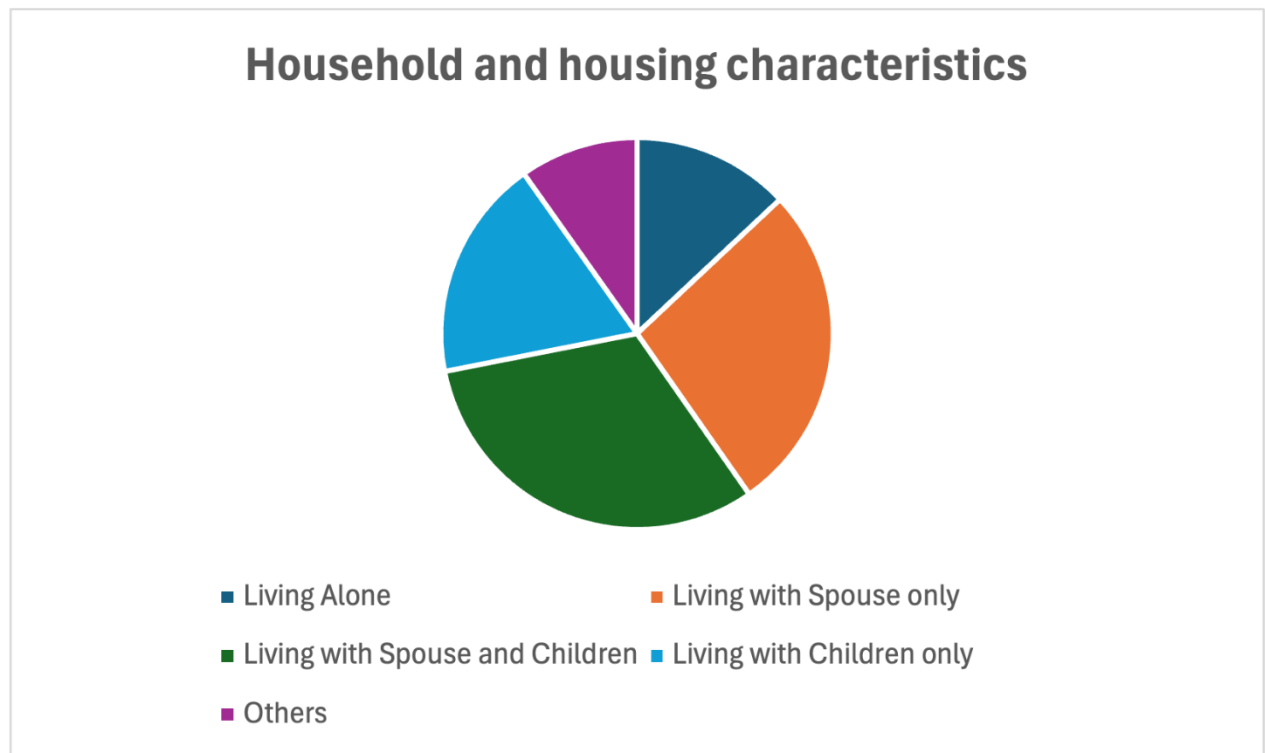


(Source: Kim et al., 2023)

Evaluation of Current Market and Industry Needs

Older adults in Hong Kong face multiple, interrelated barriers when using HA Go and comparable digital health platforms. These barriers can be grouped into operational (daily use) and onboarding (initial learning) difficulties.

Among the most frustrating barriers, standing apart from operational and onboarding difficulties, is biometric authentication. Fingerprint recognition regularly fails owing to dry, cracked, or thinned epidermal ridges, conditions frequently worsened by repeated hand-washing and household chores involving water or detergents (Luo et al., 2023). With 13% of those aged 65 and above living alone and 27.3% residing solely with a spouse (Census and Statistics Department, 2023), most continue to perform these chores without assistance, leading to persistently impaired fingerprint readability. Each failed login attempt heightens feelings of helplessness and technological rejection, converting a basic access procedure into a repeated emotional obstacle.

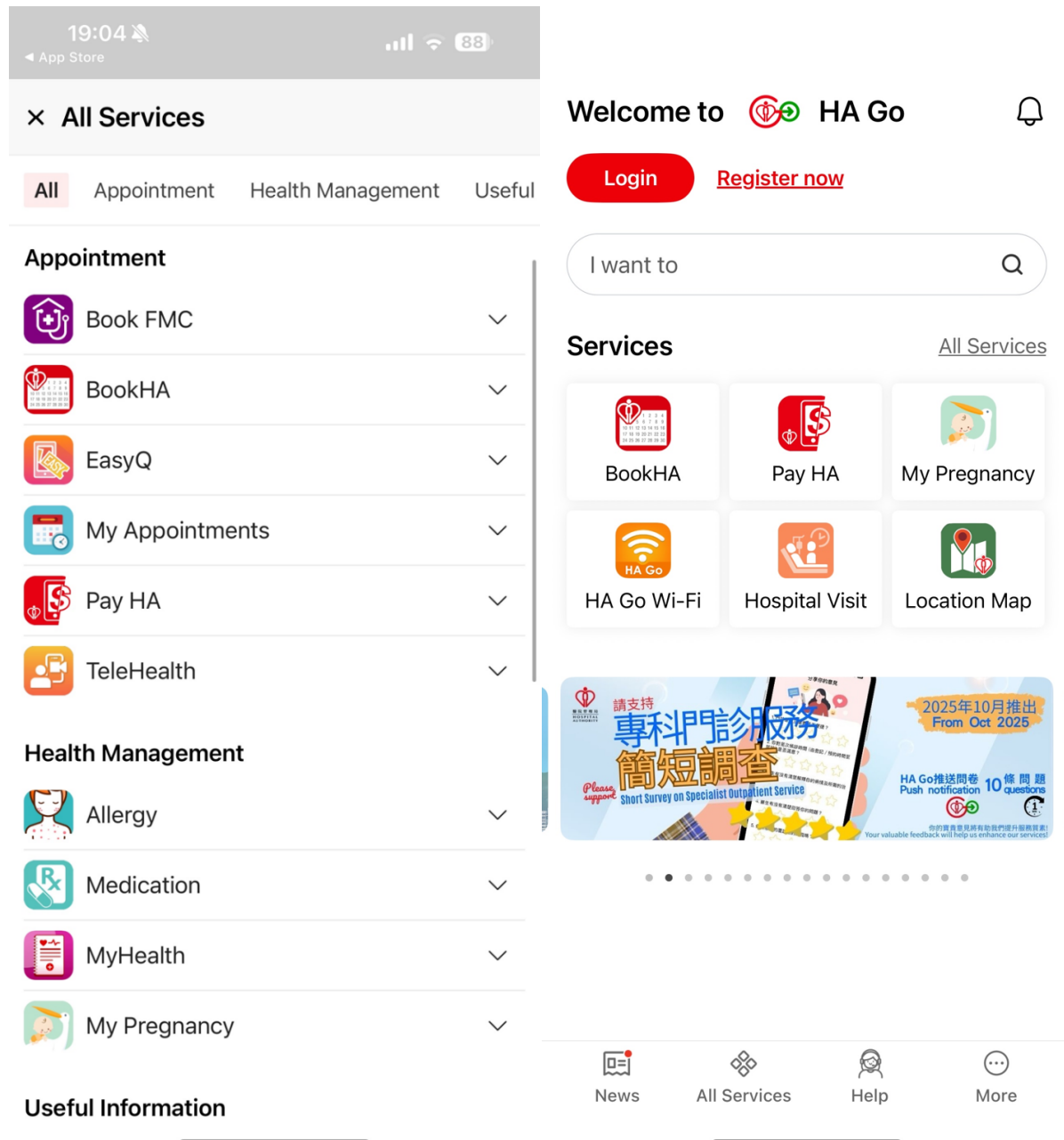


(Source: Census and Statistics Department, 2023)

The interface design of HA Go further amplifies discomfort. The application displays numerous small touch targets, densely nested menus, functionally extraneous modules (e.g., pregnancy-related services) (Hospital Authority, 2025), and occasional English-only prompts (Luo et al., 2023). Research shows that 92.5% of grassroots older adults experience general difficulty with smartphone use, while 57.6% specifically report problems with even the highly simplified LeaveHomeSafe application, which requires only a single QR-code scan (Society for Community Organization et al., 2023). That a deliberately minimal public-health tool still provokes considerable discomfort in nearly six out of ten seniors highlights the prohibitive complexity of HA Go's far more elaborate environment.



(Source: ITRC, HKCSS, 2020)



(HA Go, Version 2.34.0, Hospital Authority, 2025)

Cognitive ageing presents a further barrier, as age-related declines in memory, attention, and processing speed make learning and navigating multi-step digital interfaces profoundly challenging (Wildenbos et al., 2018).

These combined physical, interface, and cognitive barriers transform HA Go from a tool of convenience into a source of stress, demanding a senior-centric redesign.

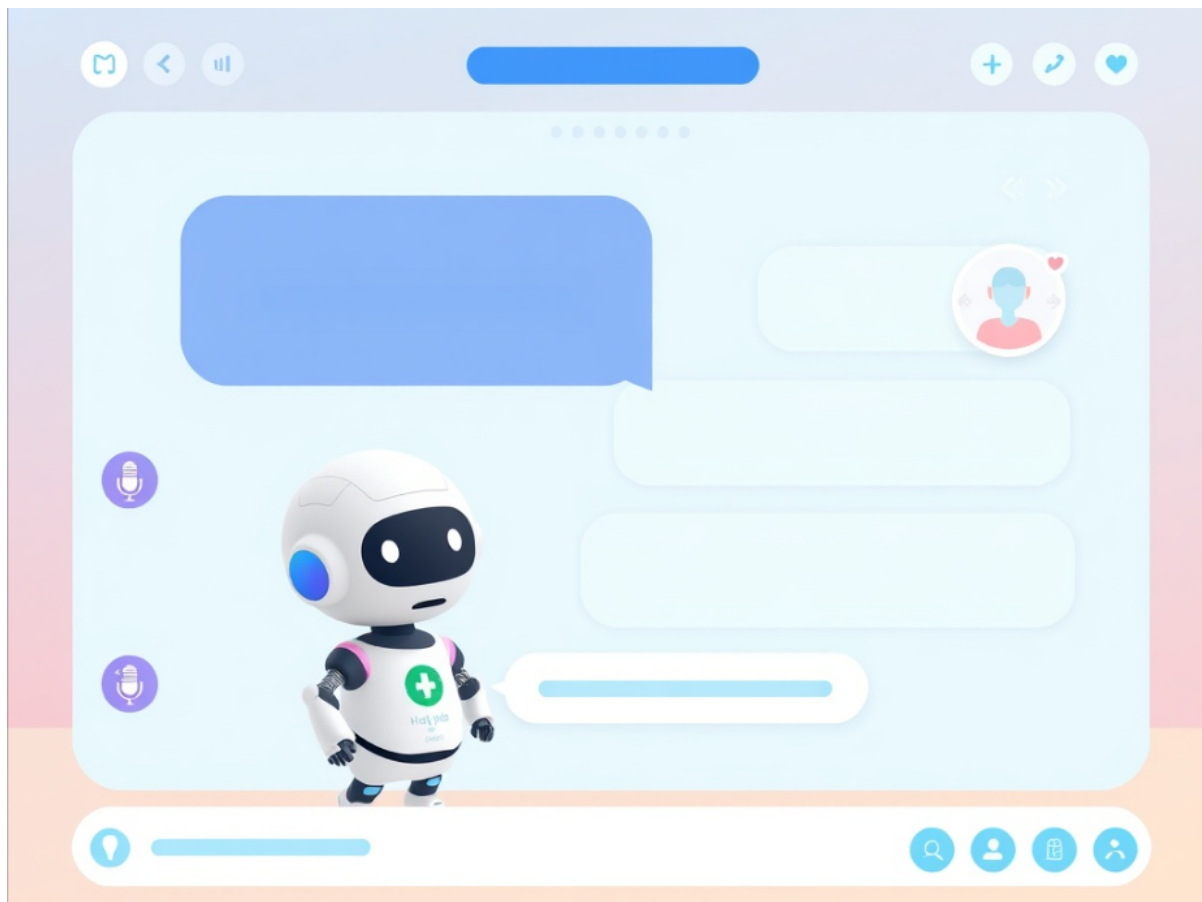
Proposing a Solution

“Your 24/7 Healthcare Chatbot” is a voice-first, AI-powered conversational assistant integrated directly into the existing HA Go app, requiring no separate download. It is designed with senior-centred features that include Cantonese-first natural voice conversation delivered with phone-call-like projection and accompanied by large-font text backup, one-stop guidance for queries such as “Show me tomorrow’s appointment,” “When do I take my blood pressure pill?” or “Explain ‘hypertension’ in simple terms,” proactive reminders for appointments and medications, plain-language translation of medical reports and jargon, a dedicated caregiver and healthcare-provider settings page for remote configuration and monitoring, and full end-to-end encryption with HA-compliant data handling.

By replacing taps and swipes with familiar spoken dialogue, the Chatbot directly addresses every major barrier identified: physical limitations are overcome by eliminating the need for precise tapping or biometric authentication, UI complexity is removed through the complete avoidance of nested menus, cognitive load is reduced because conversational context eliminates the requirement to remember multiple steps, and lack of support is resolved by providing 24/7 availability with gentle, patient repetition.

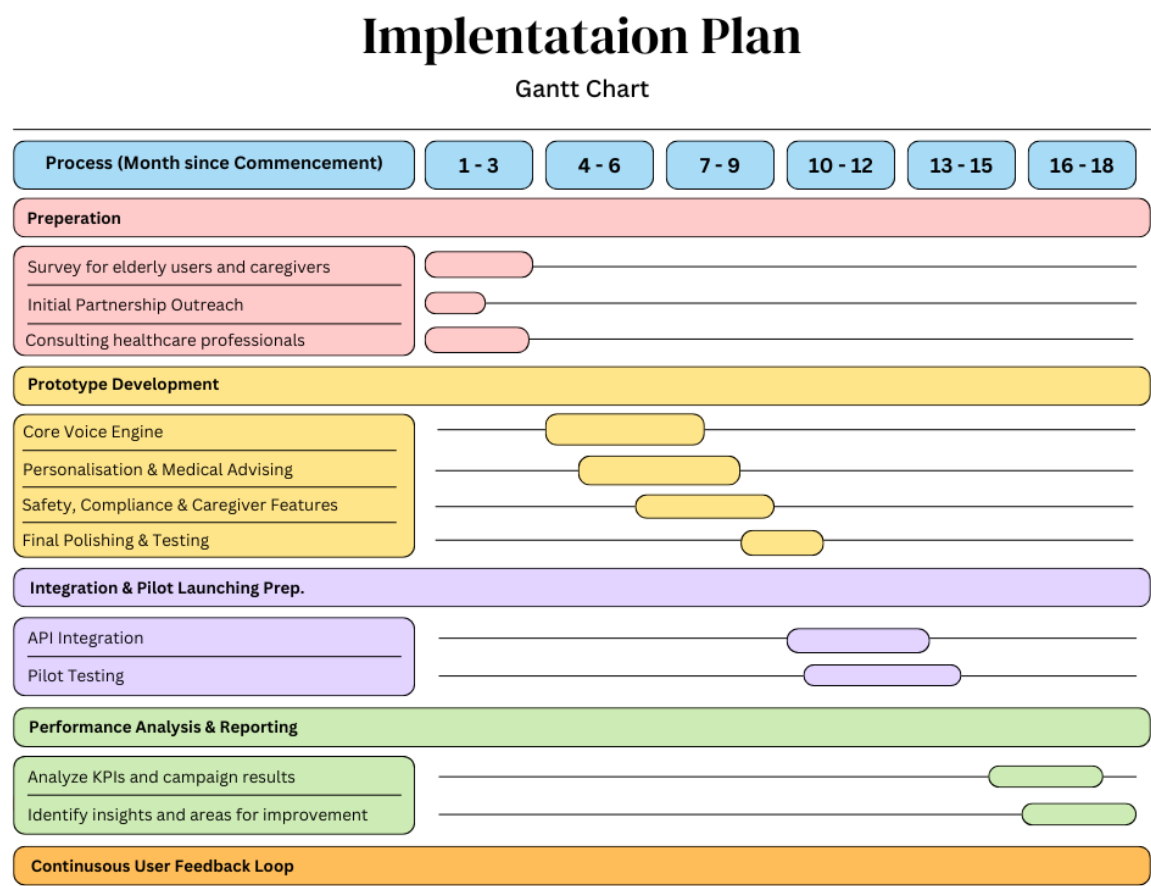
Within the TRAM framework, large fonts and voice interaction dramatically increase perceived ease of use, while reliable reminders and jargon explanation enhance perceived usefulness. At the same time, a warm, human-like tone significantly reduces discomfort and insecurity, the two inhibitors most detrimental to senior adoption.

Precedents are encouraging: the Meela AI elderly companion has received glowing feedback (for example, “It’s amazing... It’s pleasant” – Duncan, aged 89) (Meela.ai, 2025), and DBS Bank’s Joy AI demonstrates that conversational interfaces can achieve high satisfaction among older users when appropriately designed (DBS Foundation, 2025). Integration as an HA Go module aligns with the Hospital Authority’s stated openness to public–private collaboration , as expressed at the Asia Pacific Electronic Health Records Conference in 2019 (eHealth, 2020).



(Proposed User Interface of Your 24/7 Healthcare Chatbot)

Implementation Plan



Preparation

1. Survey for Elderly Users and Caregivers:
Conduct surveys and focus groups with 100+ seniors and caregivers to validate features and identify unmet needs
2. Initial Partnership Outreach:
Contact Hospital Authority, Tung Wah Group, and government Smart City/PPP (Hospital Authority, 2025) with prototype demos and user testimonials to secure MoUs, pilot commitments, and potential funding or in-kind support.
3. Consulting Healthcare Professionals
Engage geriatricians and nurses for review of medical advising accuracy, clinical workflow fit, and recommendations on safe emergency escalation protocols.

Prototype Development

1. Core Voice Engine

Develop ElevenLabs-powered Cantonese voice with Hong Kong tone (ElevenLabs, 2025), wake-word activation, large-font text fallback, customizable voice personas, and proactive notification system optimised for elderly speech patterns and background noise.

2. Personalization & Medical Advising

Implement personalised memory, proactive reminders, plain-Cantonese medical jargon translation, and context-aware responses tailored to each user's health profile and conversation history.

3. Safety, Compliance & Caregiver Features

Build HIPAA-grade encryption (U.S. Department of Health and Human Services, 2024), zero raw-audio storage, caregiver portal with conversation logs and alerts, emergency escalation triggers, and full regulatory compliance for HA Go integration and protected health data handling.

Integration & Pilot Launching Preparation

1. API Integration

Seamlessly connect to HA Go APIs for real-time access to appointments, medication lists, lab results, and booking functions while maintaining white-label branding and Hospital Authority security standards.

2. Pilot Testing

Deploy to 300–500 seniors, collect daily usage data and feedback, run bi-weekly iterations, and achieve >85% voice-task success rate and >40% weekly active usage before full rollout.

Performance Analysis & Reporting

1. Analyze KPIs and Campaign Results

Track adoption rate, task completion, missed-appointment reduction, caregiver burden relief, NPS, and sentiment improvement; produce monthly dashboards and impact reports for HA and funders.

2. Identify Insights and Areas for Improvement

Conduct post-pilot debriefs and data analysis to uncover new feature opportunities, accessibility gaps, and scalability challenges, informing Version 2.0 roadmap and long-term sustainability strategy.

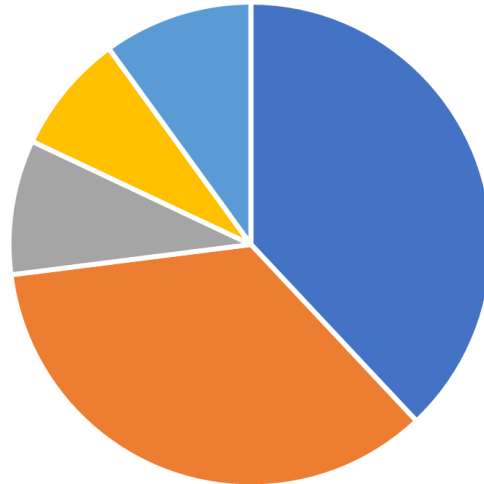
Continuous User Feedback Loop

1. Maintain permanent feedback channels (in-app voice feedback, quarterly surveys, community centre visits) throughout the entire project lifetime to ensure the chatbot keeps evolving with real elderly and caregiver needs.

Financial Plan

Budget allocation of 2M HKDs

- Personnel & Professional Services (38%)
- Technology Development & Platform Costs (35%)
- Pilot Testing & User Engagement (9%)
- Operations, Admin, Transport (8%)
- Contingency Fund (10%)



1. Personnel & Professional Services — HKD 760,000

- Full-stack Developer — 480,000 HKD
Develops and integrates all platform components, including the HA Go API and caregiver portal, and ensures reliability through QA testing.
- Project Manager — 280,000 HKD
Oversees project milestones, compliance, data analysis, reporting, and partner coordination, ensuring pilot aligns with clinical, regulatory, and operational standards.
- Healthcare Consultant Panel — 100,000 HKD
Provides clinical review, workflow validation, and emergency protocol guidance to ensure platform safety and effectiveness for elderly users.

2. Technology Development & Platform Costs — HKD 700,000

- ElevenLabs Cantonese Subscription — 103,000 HKD
Provides high-quality Cantonese TTS with wake-word activation and fallback text, enabling natural, elderly-friendly voice interactions without in-house AI engineering. (ElevenLabs, 2025)
- Cloud Infrastructure — 280,000 HKD
Hosts platform securely, providing monitoring, redundancy, and scaling capacity to support pilot deployment and future expansion.
- Emergency & Caregiver Features — 110,000 HKD
Develops alert systems, logs, and escalation triggers, ensuring caregivers and medical personnel are notified promptly of urgent health needs.
- Security Pen-testing & Audit — 27,000 HKD
Validates cybersecurity, penetration testing, and compliance to protect sensitive user health data during HA Go integration.

3. Pilot Testing & User Engagement — HKD 180,000

- Elderly Incentives (300–500 users) — 120,000 HKD
Provides vouchers for transport or meals to ensure participation and consistent engagement during the pilot phase.
- Feedback Collection — 60,000 HKD
Surveys, interviews, and transcriptions gather actionable insights on usability, satisfaction, and areas for platform improvement.

4. Operations, Admin, Transport — HKD 160,000

- Staff Travel — 40,000 HKD

Covers transport for home visits, fieldwork, and pilot support to ensure smooth project execution.

- Admin & Insurance — 80,000 HKD

Supports office operations, coordination, and liability coverage to maintain compliance and operational safety.

- Miscellaneous — 40,000 HKD

Provides for printing, consumables, and minor materials needed during project execution.

5. Contingency Fund — HKD 200,000

- Provides a buffer for unexpected technical, operational, or minor budgetary challenges to maintain project stability and continuity.

Conclusion

In summary, the current HA Go interface creates significant barriers for seniors. Our proposed chatbot offers a timely, voice-first solution to convert this source of frustration into an empowering tool, ensuring the digital healthcare ecosystem includes all of Hong Kong's aging population.

(Total Word: 1599)

Reference List

1. Lau, S. Y., & Fong, Y. F. (2021). Technology Adoption among the Elderly – Smart Older Adults in the New Decades. Hong Kong Laureate Forum.
<https://www.hklaureateforum.org/en/technology-adoption-among-the-elderly-smart-older-adults-in-the-new-decades>
2. Meela.ai. (2025). Meela.ai. <https://www.meela.ai/>
3. Kim, S., Chow, B. C., Park, S., & Liu, H. (2023). The usage of digital health technology among older adults in Hong Kong and the role of technology readiness and eHealth literacy: Path analysis. Journal of Medical Internet Research, 25, Article e41915. <https://doi.org/10.2196/41915>
4. Luo, H., Mo, B., & Li, B. (2023, December 4). Digital health literacy: The “life hurdle” to elderly Hongkongers in a smartphone-essential society. DataStory, Hong Kong Baptist University. https://datastory.hkbu.edu.hk/tp_society/digital-health-literacy-the-life-hurdle-to-elderly-hongkongers-in-a-smartphone-essential-society/
5. Census and Statistics Department. (2023, February). 2021 population census: Thematic report: Older persons. Hong Kong Special Administrative Region Government. <https://www.census2021.gov.hk/doc/pub/21c-older-persons.pdf>
6. Hospital Authority. (2025). HA Go (Version 2.34.0) [Mobile app]. Apple App Store. <https://apps.apple.com/hk/app/ha-go/id1469340861>
7. Luo, H., Mo, B., & Li, B. (2023, December 4). Digital health literacy: The “life hurdle” to elderly Hongkongers in a smartphone-essential society. DataStory, Hong Kong Baptist University. https://datastory.hkbu.edu.hk/tp_society/digital-health-literacy-the-life-hurdle-to-elderly-hongkongers-in-a-smartphone-essential-society/
8. Hong Kong Community Organization, Elderly Rights League (H.K.), Department of Social Work, The Chinese University of Hong Kong. (2023, August 20). Grassroots elderly household needs research [Press release].
<https://soco.org.hk/wp-content/uploads/2014/10/基層長者戶需要研究-新聞稿-20230820.pdf>

9. ITRC, HKCSS. (2020). *LeaveHomeSafe feature image* [Image]. ITRC, The Hong Kong Council of Social Service. <https://itrc.hkcass.org.hk/wp-content/uploads/2020/12/LeaveHomeSafeFeatureImage-e1606810760177.jpg>
10. Wildenbos, G. A., Peute, L., & Jaspers, M. W. M. (2018). Aging barriers influencing mobile health usability for older adults: A literature based framework (MOLD-US). *International Journal of Medical Informatics*, 114, 66–75. <https://doi.org/10.1016/j.ijmedinf.2018.03.012>
11. DBS Foundation. (2025, August 4). *DBS Foundation, with government dollar-for-dollar matching, contributes SGD 3 million to further accelerate digital inclusivity and increase AI literacy*. DBS. https://www.dbs.com/newsroom/DBS_Foundation_with_government_dollar_for_dollar_matching_contributes_SGD_3_million_to_further_accelerate_digital_inclusivity_and_increase_AI_literacy
12. eHealth. (2020, June 20). *eHealth News 20* [PDF]. https://www.ehealth.gov.hk/en/whats-new/ehealth-news/ehealth_news_20/ehealth_news20.pdf
13. Hospital Authority. (2025). *Public-Private Partnership (PPP) Programmes*. Hospital Authority. <https://www4.ha.org.hk/ppp/en>
14. ElevenLabs. (2025). *Free Cantonese speech to text (Scribe)*. ElevenLabs. <https://elevenlabs.io/speech-to-text/cantonese>
15. U.S. Department of Health and Human Services. (2024, July 19). *HIPAA for professionals: Laws & regulations*. <https://www.hhs.gov/hipaa/for-professionals/security/laws-regulations/index.html>